



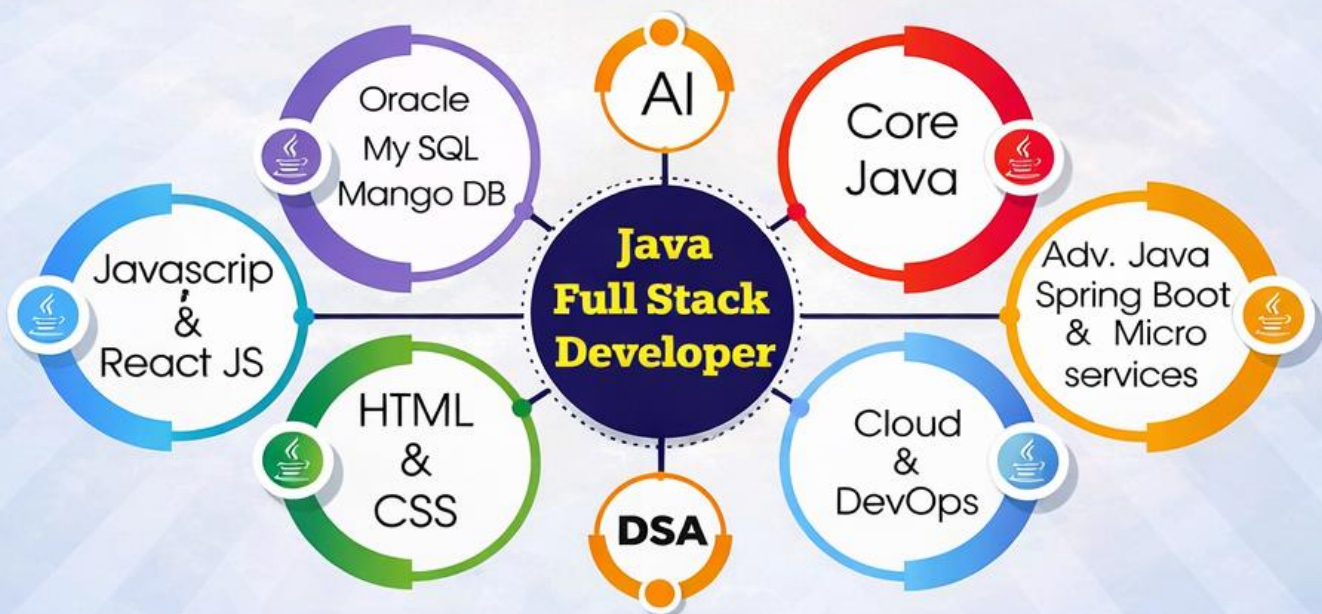
EVOLVE WEB SKILLS

Elite IT Training Institute in Solapur



FULLSTACK JAVA

with SpringBoot + React + AWS + DevOps



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JAVA FULL STACK DEVELOPMENT – SYLLABUS

1. Introduction

- What is Frontend Development
 - Frontend Technologies
 - What is Backend Development
 - Backend Technologies
 - What is Full Stack Development
 - Roles of Full Stack Developer
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2. Core Java

- What is Programming Language
- Types of Programming Languages
- Java Introduction & History
- Principles of Java
- Java Installation
- Program Execution Flow
- JDK vs JRE vs JVM
- JVM Architecture
- Identifiers & Keywords
- Variables & Data Types
- Methods
- Arrays & Strings Handling
- Object-Oriented Programming (OOPs)
- Exception Handling
- Multithreading
- File I/O Operations
- Wrapper Classes

- Autoboxing & Unboxing
-

3. Advanced Java

- JDBC
- Servlets
- JSP
- MVC Architecture
- Project Development
- Debugging

Modern Java Features

- Collections Framework
 - Lambda Expressions
 - Functional Interfaces
 - Default & Static Methods
 - Stream API
 - Method References
 - Date & Time API
 - Java 9, 11, 17, 21 Features
 - Working with IDE
 - Logical Programming
-

4. Spring Boot & Microservices

Spring Framework

- Spring Architecture
- Modules Overview
- IoC Container
- Dependency Injection
- Bean Lifecycle & Scopes

- Autowiring

Spring Boot

- Spring vs Spring Boot
- Spring Boot Introduction
- Starter POMs
- Auto Configuration
- Embedded Servers
- Actuators
- Properties vs YML

Spring Data JPA

- ORM Concepts
- Entities & Repositories
- CRUD Operations
- Custom Queries
- Pagination & Sorting
- Transactions

Spring Web MVC

- Dispatcher Servlet
- Controllers & Views
- Thymeleaf
- Forms & Validations
- Email & Logging

5. Spring REST & Microservices

- REST Architecture & Principles
- HTTP Protocol & Methods
- JSON & Jackson API
- REST API Development

- Request & Response Handling
- Postman & Swagger
- Exception Handling

Microservices

- Monolith vs Microservices
- Service Registry
- API Gateway
- Load Balancing
- Feign Client
- Kafka & Redis Integration

Security

- Authentication & Authorization
 - OAuth 2.0
 - JWT
-

6. Database (SQL)

- DBMS Introduction
 - SQL Commands (DDL, DML, DQL)
 - Constraints
 - Joins & Subqueries
 - SQL Functions
 - Procedures & Functions
-

7. Frontend Development

HTML

- Structure & Elements
- Forms & Tables
- Lists & Links

- HTML5 Semantics

CSS

- Selectors
- Box Model
- Flexbox
- Responsive Design

Bootstrap

- Grid System
 - Components (Navbar, Cards, Forms, etc.)
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8. JavaScript

- Basics & Syntax
 - Variables & Data Types
 - Arrays & Objects
 - DOM Manipulation
 - Events & Forms
 - ES6 Concepts
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9. React JS

- Introduction to React
 - Components & Props
 - Hooks & State Management
 - Context API
 - Routing
 - Axios
 - Redux
 - Testing & Deployment
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10. AWS Cloud

- Cloud Basics
 - EC2, S3, RDS
 - IAM
 - Load Balancer
 - Auto Scaling
 - AWS Lambda
 - Route 53
 - Deployment
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11. DevOps Tools

- Maven
 - GitHub
 - SonarQube
 - Docker
 - Kubernetes
 - Jenkins
 - ELK Stack
 - JIRA
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12. Additional Topics

- Data Structures & Algorithms (DSA)
- AI Basics